

DATA & AI · SMART COMPUTING · KYUNGDONG UNIVERSITY

Cheruiyot Leon Kipkorir

I build machine learning systems and then check whether their results hold up. My research looks past headline accuracy to how models behave on real, time-ordered data, and why they give the answers they do.

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github.com/scupid-90 Sokcho, South Korea

AVAILABLE 21 Dec 2026 – 26 Feb 2027, then full-time from mid-June 2027

2	6.3M	3.6 / 4.5	3
papers published, KICS Fall Conference 2026	transactions analysed in thesis research	GPA, B.S. Smart Computing (Aug 2027)	languages: English, Swahili, Korean (learning)

Research

Two published papers from my final year, both about measuring AI systems honestly.

PUBLISHED · KICS FALL CONFERENCE 2026 Undergraduate thesis Advisor: Dr. Ahmed Absi

Beyond near-perfect accuracy: fraud detection that holds up over time

Beyond Near-Perfect Accuracy: An Evaluation of Temporally Validated and Explainable Machine Learning for Mobile Money Fraud Detection

PROBLEM	Fraud models on mobile money data often report near-perfect scores. Those scores usually come from random train/test splits, which let a model learn from the future and overstate how it will perform once deployed.
WHAT I DID	Benchmarked Logistic Regression, Random Forest, XGBoost and Isolation Forest on about 6.3 million PaySim transactions, across four ways of handling class imbalance (baseline, undersampling, SMOTE, class weighting). I re-tested every model on time-ordered splits, ran feature-ablation experiments, and used SHAP across several training seeds to explain each prediction.
RESULT	Showed that headline accuracy overstates real-world performance, identified which models stay reliable under time-ordered testing, and traced which account-balance features drive their decisions.
TOOLS	Python · pandas · scikit-learn · XGBoost · imbalanced-learn · SHAP · precision, recall, F1, ROC-AUC, PR-AUC



Random split. Test data (dark) is mixed through time, so the model trains on the future.



Time-ordered split. Train on the past, test on the future, as a deployed model would.

PUBLISHED · KICS FALL CONFERENCE 2026 Seminar on AI · Fall 2026

Same candidate, different words: measuring bias in an LLM

Same Candidate, Different Words: A Name-Substitution Framework for Measuring and Mitigating Racial and

Gender Bias in Claude's Outputs

PROBLEM	Generative AI is used to write about people, but it is hard to tell whether its tone or judgement changes with a person's name.
WHAT I DID	Designed a name-substitution framework: identical prompts about the same candidate, with only the name changed to signal different racial and gender groups. I compared the outputs case by case and tested mitigation prompts in further rounds.
RESULT	A reusable, controlled method for measuring racial and gender bias in LLM outputs, published as a conference paper.
TOOLS	Claude · prompt design · controlled experiments · output analysis

Projects

Things I built and delivered for real users, a class or a team.

TEAM PROJECT · FALL 2026 Data Science course · 7 members

Restaurant sales analysis and demand prediction

PROBLEM	Restaurants waste food when they over-prepare and lose sales when they under-prepare.
WHAT I DID	Built the analysis notebook and dashboard for the team: analysed sales patterns in a Kaggle restaurant dataset and predicted customer demand.
RESULT	Presented recommendations to the class for cutting food waste and maximising sales.
TOOLS	Python · pandas · scikit-learn · Jupyter · PowerPoint

PERSONAL PROJECT Web app · Claude API

Korean Quest

A gamified TOPIK I vocabulary app with 567 words, spaced-repetition flashcards, several quiz modes and streak tracking. It uses the Claude API to read Korean text from the device camera.

TEAM PROJECT · 2026

ESP32 · C/C++ · Google Sheets

RFID smart attendance system

An ESP32 system that reads RFID cards and logs check-ins and check-outs to Google Sheets over Wi-Fi, with LCD and buzzer feedback. I also produced the reports, operator manual, presentation and promotional video.

Education and experience

Aug 2023 – Aug 2027	**B.S. Smart Computing, Kyungdong University Global Campus** GPA 3.6 / 4.5 · Data Science, Data Mining, Machine Learning, Seminar on AI, Cloud Computing, Embedded Systems
2026 – present	**Undergraduate Researcher, Kyungdong University** Mobile money fraud detection thesis; defense December 2026
Jul 2024 – present	**Part-time Staff, Daseung Supply** Shop operations in a Korean workplace, working in Korean and English; I also plan the weekday transport schedule for 7 coworkers

Skills and languages

Programming	Python, SQL, C/C++, C#, JavaScript, HTML/CSS
Data and ML	pandas, NumPy, scikit-learn, XGBoost, SHAP, imbalanced-learn, Matplotlib, Jupyter
Methods	Classification, anomaly detection, regression, clustering, temporal validation, feature ablation, model explainability
AI tools	Claude API, prompt design, LLM output evaluation
Other	Git/GitHub, Excel, PowerPoint, Google Sheets, Linux, Unity, ESP32/Arduino
Languages	English (fluent) · Swahili (fluent) · Korean (TOPIK I Level 1; KIIP Stage 2 completed)